

PHY 614-2011

Problem Set #9

Problem 1

$$(1) x = r \cos \theta \quad y = r \sin \theta \Rightarrow r^2 = x^2 + y^2$$

$$\tan \theta = \frac{y}{x}$$

$$\Rightarrow \sec^2 \theta d\theta = \frac{dy}{x} - y \frac{dx}{x^2}$$

$$\frac{r^2}{x^2} d\theta = \frac{dy}{x} - y \frac{dx}{x^2}$$

$$d\theta = \frac{x}{r^2} dy - \frac{y}{r^2} dx = \frac{1}{r} (\cos \theta dy - \sin \theta dx)$$

$$\Rightarrow$$

$$\frac{\partial \theta}{\partial x} = -\frac{1}{r} \sin \theta$$

$$\frac{\partial \theta}{\partial y} = \frac{1}{r} \cos \theta$$

$$\frac{\partial r}{\partial x} = \frac{x}{r} = \cos \theta$$

$$\frac{\partial r}{\partial y} = \frac{y}{r} = \sin \theta$$

Thus

$$\partial_x = -\frac{1}{r} \sin \theta \partial_\theta + \cos \theta \partial_r$$

$$\partial_y = \frac{1}{r} \cos \theta \partial_\theta + \sin \theta \partial_r$$

$$(\partial_x^2 + \partial_y^2) \psi = \left(\cos \theta \partial_r - \frac{1}{r} \sin \theta \partial_\theta \right) \left(\cos \theta \partial_r \psi - \frac{1}{r} \sin \theta \partial_\theta \psi \right)$$

$$+ \left(\sin \theta \partial_r + \frac{1}{r} \cos \theta \partial_\theta \right) \left(\sin \theta \partial_r \psi + \frac{1}{r} \cos \theta \partial_\theta \psi \right)$$

$$= \partial_r^2 \psi + \frac{1}{r} \partial_r \psi + \frac{1}{r^2} \partial_\theta^2 \psi$$

which yields eqn (1).

(2) Simultaneous eigenstates of H and $-i\hbar \partial_\theta$ are of the form

$$\psi_{E,m}(r, \theta) = \chi_m(r) e^{im\theta}$$

Substituting in (1) leads to

$$-\frac{\hbar^2}{2\mu} \left[\frac{1}{r} \partial_r (r \partial_r \chi_m) - \frac{m^2}{r^2} \chi_m \right] + V(r) \chi_m = E \chi_m$$

$\Rightarrow$

$$-\frac{\hbar^2}{2\mu} \left[\frac{1}{r} \partial_r (r \partial_r \chi_m) \right] + \left(V(r) + \frac{m^2 \hbar^2}{2\mu r^2} \right) \chi_m = E \chi_m$$

Define a new wave function by

$$\chi_m = \frac{1}{r^\alpha} U_m(r)$$

This gives

$$\begin{aligned} \frac{1}{r} \partial_r (r \partial_r \chi_m) &= \frac{1}{r^\alpha} U_m'' - \frac{2\alpha-1}{r^{\alpha+1}} U' \\ &+ \frac{1}{r^\alpha} \frac{\alpha^2}{r^2} U \end{aligned}$$

The term U' cancels when

$$\alpha = 1/2$$

Thus we get a 1d Schrödinger equation of the form

$$-\frac{\hbar^2}{2\mu} U_m'' + \left(V(r) + \frac{m^2 \hbar^2}{2\mu r^2} - \frac{\hbar^2}{2\mu r^2} \frac{1}{4} \right) U_m = E U_m$$

Thus the effective potential is

$$V_{\text{eff}} = V(r) + \frac{\hbar^2}{2\mu r^2} \left(m^2 - \frac{1}{4} \right)$$

Problem 2 (12.3.7)

(1) In polar coordinates

$$H = -\frac{\hbar^2}{2\mu} \left(\frac{\partial^2}{\partial \rho^2} + \frac{1}{\rho} \frac{\partial}{\partial \rho} + \frac{1}{\rho^2} \frac{\partial^2}{\partial \phi^2} \right) + \frac{1}{2} \mu \omega^2 \rho^2$$

Clearly

$$[H, \partial_\phi] = 0 \Rightarrow [H, L_z] = 0$$

Acting on $\psi_m(\rho, \phi) = R_m(\rho) e^{im\phi}$
the radial eqn is

$$\begin{aligned} -\frac{\hbar^2}{2\mu} \left(\frac{d^2}{d\rho^2} + \frac{1}{\rho} \frac{d}{d\rho} - \frac{m^2}{\rho^2} \right) R_m(\rho) \\ + \frac{1}{2} \mu \omega^2 \rho^2 R_m(\rho) \\ = E R_m(\rho) \end{aligned}$$

(2) Near $\rho \sim 0$ we may ignore both the potential term as well as the RHS and get

$$\rho \rightarrow 0, \quad \left(\frac{d^2}{d\rho^2} + \frac{1}{\rho} \frac{d}{d\rho} - \frac{m^2}{\rho^2} \right) R_m(\rho) = 0$$

The solutions must be of the form $R_m(\rho) \sim \rho^\alpha$

since the eqn is invariant under scaling $\rho \rightarrow \lambda \rho$

Then

$$\alpha(\alpha-1) + \alpha - m^2 = 0$$

$$\Rightarrow \alpha = \pm m$$

$$\Rightarrow R_m(\rho) \sim \rho^{\pm m} \quad \text{as } \rho \rightarrow 0$$

Since R should not diverge, $R_m(\rho) \sim \rho^{|m|}$

(3) As $\rho \rightarrow \infty$ we may ignore the $E \approx m^2/\rho^2$ term. The eqn then becomes

$$-\frac{\hbar^2}{2\mu} \frac{1}{\rho} \frac{d}{d\rho} \left(\rho \frac{d}{d\rho} \right) R + \frac{1}{2} \mu \omega^2 \rho^2 R = 0$$

In fact the term $\frac{1}{\rho} \frac{dR}{d\rho}$ can be also ignored, provided $R'(\rho)$ does not blow up at $\rho \rightarrow \infty$, as we will see.

$$\text{If } f = e^{-\mu \omega \rho^2 / 2\hbar}$$

$$f'' = \frac{\mu \omega^2 \rho^2}{\hbar^2} f - \frac{\mu \omega}{\hbar} f$$

$$\approx \frac{\mu \omega^2 \rho^2}{\hbar^2} f \quad \text{for } \rho \rightarrow \infty$$

which satisfies the eqn of motion

(4) Introducing

$$\epsilon = \frac{E}{\hbar\omega} \quad y = \left(\frac{\mu\omega}{\hbar}\right)^{1/2} \rho$$

we get

$$-\frac{1}{2} \left(\frac{d^2}{dy^2} + \frac{1}{y} \frac{d}{dy} - \frac{m^2}{y^2} \right) R_{Em}(y) + \frac{1}{2} y^2 R_{Em}(y) = \epsilon R_{Em}(y)$$

(5) Ansatz: $R_{Em}(y) = y^{|m|} e^{-y^2/2} U_{Em}(y)$
leads to

$$U'' + \left(\frac{2|m|+1}{y} - 2y \right) U' + 2(\epsilon - 2|m| - 2)U = 0$$

(6) let $U = \sum_{r=0}^{\infty} C_r y^r$
Then

$$\sum_{r=0}^{\infty} \left[r(r-1) C_r y^{r-2} + (2|m|+1) r C_r y^{r-2} - 2r C_r y^r + 2(\epsilon - 2|m| - 2) C_r y^r \right] = 0$$

Change summation index $r \rightarrow r-2$ in first terms
to get the recursion relation

$$\begin{aligned} & [(r+2)(r+1) + (r+2)(2|m|+1)] C_{r+2} \\ & = [2r + 2(\epsilon - 2|m| - 2)] C_r \end{aligned}$$

$$\frac{C_{r+2}}{C_r} = 2 \left[\frac{r - (\epsilon - |m| - 1)}{(r+2)(2|m|+2+r)} \right]$$

(7) If the solution is well behaved at $y \rightarrow \infty$, the series must terminate, i.e. $\exists$ a r such that

$$\epsilon = (r + |m| + 1).$$

However $U(0) = 0$, since otherwise the behavior of R at $y \rightarrow 0$ is not $y^{|m|}$.

Therefore $C_r = 0$ for all odd r .
 $\Rightarrow$

$$\begin{aligned} \epsilon &= (2k + |m| + 1) \\ k &= 0, 1, 2, \dots \end{aligned}$$

(8) If $n = 2k + |m|$

$$\epsilon = (n + 1)$$

For n even $|m|$ is also even or zero.

For n odd $|m| = 1, 3, 5, \dots$

For given even n $|m| = 0, 2, 4, \dots, n$

ie $(\frac{n}{2} + 1)$ values. Thus $m = \pm |m|$

has $(n+1)$ values since $m=0$ state is complex conjugate of itself.

$\Rightarrow$ There is a $(n+1)$ fold degeneracy.

For n odd $|m| = 1, 3, 5, \dots, n \Rightarrow \frac{n+1}{2}$ values $\Rightarrow m$ has $2 \cdot \frac{n+1}{2} = (n+1)$ values

In Cartesian coordinates we have 2 decoupled oscillators

$$E = (n_1 + n_2 + 1)$$

Thus $n = n_1 + n_2$. For given n

n_1 can take values from 0 to n , and then n_2 is determined

$\Rightarrow$ degeneracy is again $(n+1)$

(9) For $n=0$, we must have $k = m = 0$

Thus wave function is

$$\psi_{n=0} = A e^{-\frac{\mu \omega p^2}{2\hbar}}$$

To normalize, require

$$2\pi |A|^2 \int_0^\infty p dp e^{-\frac{\mu \omega p^2}{2\hbar}} = 1$$

$$\Rightarrow A = \left(\frac{\hbar}{2\pi\mu\omega} \right)^{1/2} \text{ upto a phase}$$

This state is non-degenerate and is in fact the product of ground state wave functions for the 2 oscillators in x, y directions

- For $n=1$, $|m|$ can be only 1
 $\Rightarrow m = \pm 1$

$$\begin{aligned} R_{1,\pm 1} &= \rho e^{-\frac{\mu\omega\rho^2}{2\hbar}} e^{\pm i\phi} \\ &= (x \pm iy) e^{-\frac{\mu\omega}{2\hbar^2}(x^2+y^2)} \end{aligned}$$

which are clearly linear combinations of the wavefunctions for $n=1$ in Cartesian

$$(\text{Note } H_1(\xi) = 2\xi)$$

Problem 3 (12.3.8)

$$(1) \quad A_x = -\frac{B}{2}y \quad A_y = \frac{B}{2}x$$

Then clearly

$$\vec{B} = \vec{\nabla} \times \vec{A} = B \hat{k}$$

(2) Consider the Lagrangian of a particle of mass μ and charge q is

$$L = \frac{1}{2}\mu(\dot{x}^2 + \dot{y}^2 + \dot{z}^2) - \frac{q}{c}(\dot{x}A_x + \dot{y}A_y)$$

$$L = \frac{1}{2}\mu(\dot{x}^2 + \dot{y}^2 + \dot{z}^2) + \frac{qB}{2c}(y\dot{x} - x\dot{y})$$

Clearly $P_z = \frac{\partial L}{\partial \dot{z}}$ is conserved
For a given P_z equations of motion are

$$\mu\ddot{x} + \frac{qB}{2c}\dot{y} + \frac{qB}{2c}\dot{y} = 0$$

$$\mu\ddot{y} - \frac{qB}{2c}\dot{x} - \frac{qB}{2c}\dot{x} = 0$$

Thus

$$\mu \frac{d^2 x}{dt^2} = -\frac{qB}{c} \frac{dy}{dt}$$

$$\mu \frac{d^2 y}{dt^2} = \frac{qB}{c} \frac{dx}{dt}$$

Clearly

$$\mu \left(\frac{dx}{dt} \frac{d^2 x}{dt^2} + \frac{dy}{dt} \frac{d^2 y}{dt^2} \right) = 0$$

$$\Rightarrow \frac{1}{2} \mu \frac{d}{dt} \left[\left(\frac{dx}{dt} \right)^2 + \left(\frac{dy}{dt} \right)^2 \right] = 0$$

ie the quantity $v_x^2 + v_y^2$ is a constant $\Rightarrow$ Motion is in a circle. The frequency is clearly $\frac{qB}{c\mu}$, since by rescaling

$$t = \frac{\mu c}{qB} \tau$$

the equations become

$$\frac{d^2 x}{d\tau^2} = -\frac{dy}{d\tau}, \quad \frac{d^2 y}{d\tau^2} = \frac{dx}{d\tau}$$

Alternatively, in polar coord.

$$v_x = -v \sin \theta, \quad v_y = v \cos \theta$$

and the equations become

$$\mu \frac{d\theta}{dt} = \frac{qB}{c}$$

$$\Rightarrow \theta = \omega_0 t + \theta_0 \quad \omega_0 = \frac{qB}{\mu c}$$

$$\begin{aligned}
 (3) \quad [Q, P] &= \frac{1}{qB} \left[c P_x + \frac{qB y}{2}, P_y - \frac{qB x}{2c} \right] \\
 &= \frac{1}{qB} \left\{ \frac{qB}{2} [y, P_y] - \frac{qB}{2} [P_x, x] \right\} \\
 &= i
 \end{aligned}$$

Then substituting

$$H = \frac{p^2}{2\mu} + \frac{1}{2} \mu \omega_0^2 Q^2$$

where $\omega_0 = \frac{qB}{\mu c}$
 This is the hamiltonian
 of a one dim HO
 $\Rightarrow E_n = (n + \frac{1}{2}) \hbar \omega_0$

(4) In terms of P_x, x, P_y, y

$$\begin{aligned}
 H &= \frac{1}{2\mu} (P_x^2 + P_y^2) + \frac{1}{2} \mu \left(\frac{\omega_0}{2} \right)^2 (x^2 + y^2) \\
 &\quad - \frac{\omega_0}{2} (x P_y - y P_x) \\
 &= H\left(\frac{\omega_0}{2}, \mu\right) - \frac{\omega_0}{2} L_z
 \end{aligned}$$

$H(\frac{\omega_0}{2}, \mu)$ may be diagonalized by choosing simultaneous eigenstates of $H(\frac{\omega_0}{2}, \mu)$, L_z

Since

$$[H, L_z] = [H(\frac{\omega_0}{2}, \mu), L_z] - \frac{\omega_0}{2} [L_x, L_z] = 0$$

we can use the same basis to diagonalize H

Thus the eigenvalues are

$$E = \hbar \frac{\omega_0}{2} [2k + |m| + 1 - m]$$

$$\text{since } L_z |\psi\rangle = m\hbar |\psi\rangle$$

Define an integer

$$n = k + \frac{|m|}{2} - \frac{m}{2}$$

This is clearly an integer both for odd and even m

$$\Rightarrow E = \hbar \omega_0 (n + \frac{1}{2}).$$

Problem #6 (12.5.1)

$$\psi'_x(x, y) = \psi_x(x, y) + \epsilon_z y \frac{\partial \psi_x}{\partial x} - x \epsilon_z \frac{\partial \psi_x}{\partial y} - \epsilon_z \psi_y(x, y) + O(\epsilon_z^2)$$

$$\psi'_y(x, y) = \epsilon_z \psi_x(x, y) + \psi_y(x, y) + y \epsilon_z \frac{\partial \psi_y}{\partial x} - x \epsilon_z \frac{\partial \psi_y}{\partial y}$$

Rewrite this as

$$\psi'_x(x, y) = \psi_x(x, y) - \frac{i\epsilon_z}{\hbar} L_z \psi_x - \epsilon_z \psi_y$$

$$\psi'_y(x, y) = \psi_y(x, y) - \frac{i\epsilon_z}{\hbar} L_z \psi_y + \epsilon_z \psi_x$$

$$\text{where } L_z = -i\left(x \frac{\partial}{\partial y} - y \frac{\partial}{\partial x}\right)$$

the result then follows

Problem 7 (12.5.2)

$$(1) J_x^{(1/2)} = \frac{\hbar}{2} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \quad J_y^{1/2} = \frac{\hbar}{2} \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$$

$$J_z^{1/2} = \frac{\hbar}{2} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

Direct matrix multiplication
gives result

$$(2) J_x^{(1)} = \frac{\hbar}{\sqrt{2}} \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}$$

$$J_y^{(1)} = \frac{\hbar}{\sqrt{2}} \begin{pmatrix} 0 & -i & 0 \\ i & 0 & -i \\ 0 & i & 0 \end{pmatrix}$$

$$J_z^{(1)} = \hbar \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix}$$

Direct multiplication
verifies algebra

For $j=3/2$ start with

$$\langle m', 3/2 | J_+ | m, 3/2 \rangle = \sqrt{(\frac{3}{2}-m)(\frac{5}{2}+m)} \delta_{m', m+1}$$

and by definition

$$J_- = J_+^\dagger$$

The matrix for J_+ is then

$$\langle 5/2 | J_+ | 3/2 \rangle = 0 \hbar$$

$$\langle 3/2 | J_+ | 1/2 \rangle = \sqrt{3} \hbar$$

$$\langle 1/2 | J_+ | -1/2 \rangle = 2 \hbar$$

$$\langle -1/2 | J_+ | -3/2 \rangle = \sqrt{3} \hbar$$

Thus

$$J_+ = \begin{pmatrix} 0 & \sqrt{3} & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & \sqrt{3} \\ 0 & 0 & 0 & 0 \end{pmatrix} \hbar$$

$$J_- = \begin{pmatrix} 0 & 0 & 0 & 0 \\ \sqrt{3} & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & \sqrt{3} & 0 \end{pmatrix} \hbar$$

$$J_z = \begin{pmatrix} +3/2 & 0 & 0 & 0 \\ 0 & +1/2 & 0 & 0 \\ 0 & 0 & -1/2 & 0 \\ 0 & 0 & 0 & -3/2 \end{pmatrix} \hbar$$

Problem 8 (12.5.3)(1) For some given j

$$\langle j, m' | J_x | j, m \rangle = \frac{\hbar}{2} \left[\delta_{m', m+1} \sqrt{(j-m)(j+m+1)} + \delta_{m', m-1} \sqrt{(j+m)(j-m+1)} \right]$$

$$\langle j, m' | J_y | j, m \rangle = \frac{\hbar}{2i} \left[\delta_{m', m+1} \sqrt{(j-m)(j+m+1)} - \delta_{m', m-1} \sqrt{(j+m)(j-m+1)} \right]$$

Clearly $\langle j, m | J_x | j, m \rangle = \langle j, m | J_y | j, m \rangle = 0$

$$\begin{aligned} (2) \quad \langle j, m | J_x^2 | j, m \rangle &= \sum_{m'=-j}^j \langle j, m | J_x | j, m' \rangle \langle j, m' | J_x | j, m \rangle \\ &= \sum_{m'=-j}^j \langle j, m | J_x | j, m-1 \rangle \langle j, m-1 | J_x | j, m \rangle \\ &\quad + \sum_{m'=-j}^j \langle j, m | J_x | j, m+1 \rangle \langle j, m+1 | J_x | j, m \rangle \\ &= \frac{\hbar^2}{4} \left[(j-m+1)(j+m)(j+m)(j-m+1) \right. \\ &\quad \left. + (j+m+1)(j-m)(j-m)(j+m+1) \right] \\ &= \frac{\hbar^2}{2} [j(j+1) - m^2] \end{aligned}$$

By symmetry $\langle J_x^2 \rangle = \langle J_y^2 \rangle$
 In fact the result may be
 obtained by using

$$\langle J^2 \rangle = \langle J_x^2 \rangle + \langle J_y^2 \rangle + \langle J_z^2 \rangle$$

$$(3) \quad (\Delta J_x)^2 = \langle J_m | J_x^2 | J_m \rangle - \langle J_x \rangle^2$$

$$\langle J_x^2 \rangle = \langle J_y^2 \rangle$$

$$\Rightarrow (\Delta J_x)^2 (\Delta J_y)^2 = \left(\frac{\hbar^2}{2} \right)^2 [j(j+1) - m^2]^2$$

Now consider

$$\langle J_m | J_x J_y | J_m \rangle$$

$$= \langle J_m | J_x | J, m-1 \rangle \langle J, m-1 | J_y | J_m \rangle$$

$$+ \langle J_m | J_x | J, m+1 \rangle \langle J, m+1 | J_y | J_m \rangle$$

$$= -\frac{\hbar^2}{4i} (j-m+1)(j+m)$$

$$+ \frac{\hbar^2}{4i} (j+m+1)(j-m)$$

$$= -\frac{\hbar^2}{2i} m$$

Thus

$$(\Delta J_x)^2 (\Delta J_y)^2 - |\langle J_x J_y \rangle|^2$$

$$= \frac{\hbar^4}{4} \{ [j(j+1) - m(m-1)] [j(j+1) - m(m+1)] \}$$

The factor $j(j+1) - m(m+1)$ is always positive, since
 $-j \leq m \leq j$

The first factor may be simplified by writing

$$m = -j + n$$

So that

$$0 \leq n \leq 2j$$

Then

$$j(j+1) - m(m-1) = (2j-n)n + n.$$

The right hand side is now manifestly non-negative

Thus

$$(\Delta J_x)^2 (\Delta J_y)^2 \geq |\langle J_x J_y \rangle|^2$$

(4) Equality is satisfied if any one of the factors vanish

$$j(j+1) - m(m+1) = 0 \quad \text{for } m = j$$

$$j(j+1) - m(m-1) = 0 \quad \text{for } m = -j$$

$\Rightarrow$ Minimum uncertainty state is $|j, \pm j\rangle$

Problem 11 (12.5.10)

Action on wave functions

$$\Psi_{\alpha m}(\theta, \phi) = P_{\alpha}^m(\theta) e^{im\phi}$$

eigenvalue eqn for L^2 becomes

$$\left(\frac{1}{\sin\theta} \frac{\partial}{\partial\theta} \left(\sin\theta \frac{\partial}{\partial\theta} \right) + \frac{\alpha}{\hbar^2} - \frac{m^2}{\sin^2\theta} \right) P_{\alpha}^m = 0$$

$$u = \cos\theta$$

$$\frac{\partial}{\partial\theta} = -\sin\theta \frac{\partial}{\partial u} = -\sqrt{1-u^2} \frac{\partial}{\partial u}$$

$$\sin\theta \frac{\partial}{\partial\theta} = -(1-u^2) \frac{\partial}{\partial u}$$

$$\frac{\partial}{\partial\theta} \left(\sin\theta \frac{\partial}{\partial\theta} \right) = \sqrt{1-u^2} \frac{\partial}{\partial u} \left((1-u^2) \frac{\partial}{\partial u} \right)$$

Thus equation becomes

$$\left[\frac{\sqrt{1-u^2}}{\sqrt{1-u^2}} \frac{\partial}{\partial u} \left((1-u^2) \frac{\partial}{\partial u} \right) + \frac{\alpha}{\hbar^2} - \frac{m^2}{1-u^2} \right] P_{\alpha}^m = 0$$

For $m=0$ this becomes

$$(1-u^2) \frac{d^2 P_{\alpha}^0}{du^2} - 2u \frac{dP_{\alpha}^0}{du} + \left(\frac{\alpha}{\hbar^2} \right) P_{\alpha}^0 = 0$$

Consider the series solution

$$p_{\alpha}^0 = \sum_{n=0}^{\infty} C_n u^n$$

$$(p_{\alpha}^0)'' = \sum n(n-1) C_n u^{n-2}$$

Thus eqn is

$$\sum_{n=0}^{\infty} (1-u^2) n(n-1) C_n u^{n-2} - 2 \sum n C_n u^n + \frac{\alpha}{h^2} \sum C_n u^n = 0$$

or

$$\sum_{n=0}^{\infty} \left[n(n-1) C_n u^{n-2} - n(n-1) C_n u^n - 2n C_n u^n + \frac{\alpha}{h^2} C_n u^n \right] = 0$$

$$\text{Using } \sum_{n=0}^{\infty} n(n-1) C_n u^{n-2} = \sum_{n=0}^{\infty} (n+1)(n+2) C_{n+2} u^n$$

we get, equating the coeff of u^n

$$(n+1)(n+2) C_{n+2} - \left[n(n+1) - \frac{\alpha}{h^2} \right] C_n = 0$$

$$\frac{C_{n+2}}{C_n} = \frac{n(n+1) - \frac{\alpha}{h^2}}{(n+1)(n+2)}$$

Clearly as $n \rightarrow \infty$ $\frac{C_{n+2}}{C_n} \rightarrow 1$
and the series diverges.

Since $u=1 \Rightarrow \theta = 0, \pi$, and we require that the wavefunction is regular all over S^2 , the series must terminate.

Thus for a given α , there must be some n such that

$$\alpha = \hbar^2 n(n+1)$$

These are then allowed eigenvalues of L^2

When $C_1 = 0$ the eigenfunction is even $\rightarrow$ corresponds to n even

When $C_0 = 0$ n is odd and the eigenfunction is odd

• For $n=0$ $P_0 = 1/\sqrt{4\pi}$ (chosen normalization)

For $n=1$, $C_3=0$

$$P_1(u) = u \left(\frac{3}{4\pi} \right)^{1/2}$$

For $n=2$, $C_4=0$ $P_2(u) = \left(\frac{5}{16\pi} \right)^{1/2} (3u^2 - 1)$

Problem 12 (12.5.13)

$$\left. \begin{aligned} Y_1^{+1} &= -\left(\frac{3}{4\pi}\right)^{1/2} \frac{1}{\sqrt{2}r} (x + iy) \\ Y_1^{-1} &= \left(\frac{3}{4\pi}\right)^{1/2} \frac{1}{\sqrt{2}r} (x - iy) \end{aligned} \right\} \begin{array}{l} \text{Follows} \\ \text{directly} \\ \text{from eqn} \\ (12.5.39) \end{array}$$

Thus

$$Y_1^{+1} + Y_1^{-1} = \left(\frac{3}{4\pi}\right)^{1/2} \frac{1}{\sqrt{2}r} [-2iy]$$

$$-Y_1^{+1} + Y_1^{-1} = \left(\frac{3}{4\pi}\right)^{1/2} \frac{1}{\sqrt{2}r} [2x]$$

$$\Rightarrow y = i \left(\frac{4\pi}{3}\right)^{1/2} \frac{r}{\sqrt{2}} (Y_1^{+1} + Y_1^{-1})$$

$$x = \left(\frac{4\pi}{3}\right)^{1/2} \frac{r}{\sqrt{2}} (Y_1^{-1} - Y_1^{+1})$$

$$z = \left(\frac{4\pi}{3}\right)^{1/2} r Y_1^0$$

Thus the wavefunction is

$$\psi = N \left[\frac{4\pi}{3}\right]^{1/2} \left[2r Y_1^0 + \frac{r}{\sqrt{2}} (Y_1^{-1} - Y_1^{+1}) + i \frac{r}{\sqrt{2}} (Y_1^{+1} + Y_1^{-1}) \right] e^{-\alpha r}$$

or

$$\psi = N \left(\frac{4\pi}{3} \right)^{1/2} \left[2r Y_1^0 + \frac{r}{\sqrt{2}} Y_1^{-1} (1+i) + \frac{r}{\sqrt{2}} Y_1^1 (i-1) \right] e^{-\alpha r}$$

N is determined by normalizing the ψ . However since the Y_l^m 's are orthonormal,

$$P(l_z = -1) = c \frac{1}{2} |1+i|^2 = c$$

$$P(l_z = +1) = c \frac{1}{2} |i-1|^2 = c$$

$$P(l_z = 0) = c \cdot 4$$

c is determined by requiring

$$P(l_z = -1) + P(l_z = +1) + P(l_z = 0) = 1$$

$$\Rightarrow c = \frac{1}{6}$$

This gives the result